one lone pair of electrons in one tetrahedral position hence it has a pyramidal shape as shown in the figure.

3.1.5 Nitric acid

Preparation

Nitric acid is prepared by heating equal amounts of potassium or sodium nitrate with concentrated sulphuric acid.

$$KNO_3 + H_2SO_4 \longrightarrow KHSO_4 + HNO_3$$

The temperature is kept as low as possible to avoid decomposition of nitric acid. The acid condenses to a fuming liquid which is coloured brown by the presence of a little nitrogen dioxide which is formed due to the decomposition of nitric acid.

$$4HNO_3 \longrightarrow 4NO_2 + 2H_2O + O_2$$

Commercial method of preparation

Nitric acid prepared in large scales using Ostwald's process. In this method ammonia from Haber's process is mixed about 10 times of air. This mixture is preheated and passed into the catalyst chamber where they come in contact with platinum gauze. The temperature rises to about 1275 K and the metallic gauze brings about the rapid catalytic oxidation of ammonia resulting in the formation of NO, which then oxidised to nitrogen dioxide.

$$4NH_3 + 5O_2 \longrightarrow 4NO + 6H_2O + 120\ kJ$$

$$2NO + O_2 \longrightarrow 2NO_2$$

The nitrogen dioxide produced is passed through a series of adsorption towers. It reacts with water to give nitric acid. Nitric oxide formed is bleached by blowing air.

$$3NO_2 + H_2O \longrightarrow 2HNO_3 + NO$$

Properties

Pure nitric acid is colourless. It boils at 86 °C. The acid is completely miscible with water forming a constant boiling mixture (98% HNO_3, Boiling point 120.5 °C). Fuming nitric acid contains oxides of nitrogen. It decomposes on exposure to sunlight or on being heated, into nitrogen dioxide, water and oxygen.

$$4HNO_3 \longrightarrow 4NO_2 + 2H_2O + O_2$$

Due to this reaction pure acid or its concentrated solution becomes yellow on standing.

In most of the reactions, nitric acid acts as an oxidising agent. Hence the oxidation state changes from +5 to a lower one. It doesn't yield hydrogen in its reaction with metals. Nitric acid can act as an acid, an oxidizing agent and an nitrating agent.

As an acid: Like other acids it reacts with bases and basic oxides to form salts and water

$$ZnO + 2HNO_3 \longrightarrow Zn(NO_3)_2 + H_2O$$

$$3FeO + 10HNO_3 \longrightarrow 3Fe(NO_3)_3 + NO + 5\ H_2O$$

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